CROSS-DOMAIN AGENT COLLABORATION

Ontology-Guided Collaboration Among Personal Assistants

Shiyao Zhang
Xi'an Jiaotong-Liverpool University

Agent on Request
An interactive demo of personal-assistant collaboration

RESEARCH QUESTION

How can independently owned assistants coordinate a task when their concepts and disclosure boundaries differ?

01 Motivation

A large language model (LLM) can miss the user's intended meaning or acceptable sharing boundary.

Across stakeholders, assistants may model the same concept differently and follow different disclosure policies. Collaboration needs coordination of meaning and authorization.

02 Research approach

Ontology as context

Concept relations guide reasoning. A state registry records received facts and provenance.

Task-driven modular alignment

Align the concepts needed for the current task while retaining local representations.

This demo builds on prior work in cross-domain agent dialogue for ontology alignment and data exchange.

03 Collaboration framework

CONCEPTUAL RESEARCH MODEL

ALICE
Stakeholder A

Local ontology
Concepts and relations

Own data and skills

Own disclosure policy

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TASK MODULE

Meeting scheduling
Align task-relevant concepts

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BOB
Stakeholder B

Local ontology
Concepts and relations

Own data and skills

Own disclosure policy

Research model: align meaning for the task. The prototype implements configurable knowledge graphs and scoped data exchange.

PROTOTYPE EXECUTION PATH

Planner proposal — Live LLM / Controlled → **Authorize** — Owner-defined scope → **Exchange** — Approved snapshot → **Update state** — Confirmed receipt

04 Reference demonstration

1 Share	2 Extend	3 Match	4 Approve	5 Inspect
Tue-Wed free slots	Thu free slots only	Thu 15:00-15:30	Bob's owner: topic only	Confirmed received state

No initial overlap. Adding Thursday enables a match. Keeping a boundary or declining remain valid.

05 Disclosure control

The model proposes.
Permissions govern execution.

Example calendar projection

SHARED	WITHHELD
day	title
time	note
status	

Exact previews and scoped permissions restrict fields, dates, uses and expiry. Confirmed receipts determine what enters the recipient's knowledge.

Stopping sharing blocks future disclosure. Already received copies remain.

06 Research instrument

An exploratory demo with synthetic data and optional, separately consented state traces.

Next: semantic conflicts, module selection and users' understanding of disclosure boundaries.